

AI-Assisted Cryptocurrency Trading Simulator

Functional Specification

Student Name: Eamon Goonan ID Number: 19765759

Student Name: Mudiaga Dortie ID Number: 19339753

Staff Member Consulted for supervision: Dr. Hossein Javidnia

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1. Introduction

1.1 Overview

Our project is a cryptocurrency trading simulator designed to offer an immersive, risk-free trading experience for beginners and seasoned traders alike. Built primarily with the Django web application framework, this platform simulates real market conditions, allowing users to trade with mock currency while mirroring the actual cryptocurrency market dynamics.

Key features include:

1. **AI-Driven News Feed:** The most complex feature of our application is the AI-powered news feed. This feature leverages sophisticated sentiment analysis to evaluate incoming news, providing insights into how each piece of news might impact the market value of various cryptocurrencies. This allows users to make more informed trading decisions based on current market sentiments.
2. **Live Market Data:** Users gain access to real-time and historical cryptocurrency data from Binance, the world's largest crypto trading platform. This feature is designed to help users understand market trends and develop effective trading strategies.
3. **Trader Following and Analysis:** The platform enables users to follow successful traders on Binance. By analysing these experts' trades and strategies, users can learn and potentially mirror their trading patterns to improve their own performance.
4. **Detailed Trading Reports:** Each trade made by a followed trader generates a detailed report. These reports include key metrics like Return on Investment (ROI), Profit/Loss, and other relevant statistical data, aiding users in refining their trading tactics.
5. **User-Friendly Interface:** Users can easily register an account and start trading mock cryptocurrencies. The interface is designed to be intuitive, catering both to beginners and experienced traders.
6. **Risk-Free Environment:** The core aim of the simulator is to provide a safe learning environment. Users can practise and hone their trading skills without the fear of losing real money, making it an ideal platform for beginners.

Our vision is to empower users with the tools, data, and insights needed to make educated cryptocurrency investment decisions, enhancing their trading proficiency in a simulated, risk-free environment.

1.2 Business Context

This project introduces a web application designed to serve as a comprehensive platform for individuals embarking on their cryptocurrency trading journey. It targets two primary user groups: beginners seeking a risk-free environment to learn and practice, and experienced traders looking to test new strategies or benchmark their progress against peers.

The integration of an advanced AI feature for news sentiment analysis significantly enhances the application's utility. This AI module analyses real-time news, providing users with insights on how current events might influence cryptocurrency markets. This feature is especially valuable given the inherent volatility of these markets, enabling both novice and seasoned traders to make more informed decisions.

Accessibility is a key consideration; users can access the platform from various devices, including smartphones and computers, ensuring they stay informed and engaged regardless of location.

In terms of monetization, there are several potential pathways. The application could incorporate targeted advertisements, utilising the UI space efficiently. Alternatively, a subscription-based model could be introduced, offering users premium access to advanced features and AI-driven insights for a monthly fee. This strategic flexibility in monetization ensures that the platform can adapt to various business goals while continuing to provide value to its users.

1.3 Glossary

Cryptocurrency: A digital currency in which transactions are verified and records maintained by a decentralised system using cryptography, rather than by a centralised authority.

Simulator: Program that enables the operator to reproduce or represent under test conditions phenomena likely to occur in actual performance.

Binance: Worlds biggest bitcoin exchange and altcoin crypto exchange in the world by volume.

Bitcoin: A cryptocurrency that is designed to act as money and a form of payment.

Altcoin: An alternative digital currency to Bitcoin.

Long Position: What a trader has purchased when they buy a security or derivative with the expectation that it will rise in value

Short Position: Investing in such a way that the investor will profit if the value of the asset falls.

Return On Investment (ROI): ROI measures the amount of return on a particular investment, relative to the investment's cost.

Market Order: A market order is an order to buy or sell a stock at the market's current best available price

Limit Order: A limit order is an order to buy or sell a stock with a restriction on the maximum price to be paid or the minimum price to be received

2. General Description

2.1 Product / System Functions

1. **Register:** Users can create an account to access the platform's features, including AI-driven insights.
2. **Login:** Secure access to user accounts, enabling interaction with both standard and AI-enhanced features.
3. **Delete Account:** Users have the option to permanently remove their account and associated data from the platform.
4. **Follow Traders:** In addition to following successful traders from Binance, users can now leverage AI insights to better understand the impact of news events on these traders' strategies.
5. **Trade Mock Currency:** Users engage in risk-free trading of mock cryptocurrencies, with the added advantage of AI-analysed market sentiments to guide their trading decisions.
6. **Review Statistical Data:** This function is expanded to include not only traditional metrics like ROI and PnL but also AI-generated analyses. These AI insights offer a deeper understanding of how external news events influence the performance of their own trades and those of the traders they follow.

These functions collectively aim to provide a comprehensive and enriched trading simulation experience, blending traditional trading tools with innovative AI-driven market analysis.

2.2 User Characteristics and Objectives

Our intended users will be able to access and browse the internet and will be familiar with the

basics of cryptocurrency trading such as long and short positions, market and limit orders and have the ability to read basic statistics and graphs. The application will be hosted on a website and not all users will have access to all the features of the application. The permissions go as follows:

Unregistered Users:

These are unregistered or not signed in users, their permissions are limited to either registering an account or signing in to an account.

Registered Users:

These are users who have an account and they will have access to all of the features the web application provides for its members.

Administrators:

These are the maintainers of the website and they will have elevated permissions, such as being able to update the user interface and adding or removing features for the web application.

2.3 Operational Scenarios

Register:

Users can create an account on the website, inputting information such as name, username, email address and password.

Login:

Users can login to the website by inputting their registered details. This is a prerequisite for the following use cases.

Analysing AI-Assisted Newsfeed:

Users can access a live news feed that provides the latest crypto news, with an ai powered sentiment analysis on each post. This feature would be beneficial for users who struggle to understand complex news and are looking for a simple explanation of the sentiment.

Make a Trade:

Users will be provided with a set amount of mock currency. They will be able to choose a cryptocurrency to invest in from a list of the most popular crypto currencies. They will then choose the amount of crypto currency they would like to purchase.

Review User Trades:

Users will be able to select a "Portfolio" button, where they can view their trades. Users will then be able to select a particular trade from this list, should they wish to view more detailed information.

Follow a Trader:

Users will be provided with a list of suggested top Binance traders, whom they can “follow” in order to track their trades and review their trading history.

Review Followed User Trades:

Users will be able to select a “Following” button, where they can view the trades of followed users. Users will then be able to select a particular trade from this list, should they wish to view more detailed information.

Delete Account:

Users can delete their accounts by going into “Profile settings”. They will have to input their password in order to delete their account.

2.4 Constraints

Time Constraints:

In this project we are going to be using a lot of technologies unfamiliar to us so we will need to allow for some time to research within our allocated time for a task. Overall as we are under a time constraint we will have to manage our time well. To manage our time well we are going to use Agile methodology and employ weekly sprints and regular standups.

Accessibility Constraints:

As we intend to host a wide range of users, we will have to ensure that all users can perceive, understand, navigate and interact with the information on the web application. For example including large buttons, text and enough spacing for those who are visually impaired. We will consult academic UI design principles for guidance.

Binance Api Constraints:

There is a limitation to the Binance api so we will have to track traders' return on investments manually, using web scraping to do so .

Internet Constraints:

As our application will be based online, users must acquire internet access in order to use our app and its functionality.

Web Socket:

The web socket that will be used to relay the live crypto news is from <https://news.treeofalpha.com/>. It is a free web socket and provides a lot of information such as the link to the source, timestamps, images if any and associated crypto currencies.

3. Functional Requirements

Register/Login

Description: Users must register before being able to log in. For this they will navigate to the registration button which will be displayed on the homepage if the user is not logged in.

Criticality: This requirement is of high criticality. It is essential for users to register and login to use the application, as we must log their usage, i.e. their trades, available currency and followed Binance users.

Technical Issues: This data must be stored securely, in a database that is protected from attacks.

Dependencies: N/A.

Making a trade

Description: The system must check if the user has the available mock currency to make the trade. The Binance API will then provide the currency exchange rate before the trade is executed. The user's resulting currency will be updated in their portfolio.

Criticality: This requirement is of high criticality. This is a pivotal feature of the application, as our primary goal is to allow the user to trade using mock currencies. This feature will be our primary focus initially.

Technical Issues: We must implement the Binance API effectively in order to ensure trades are executed at the correct exchange rate.

Dependencies: Users must be logged in before they can make trades.

Provisions of trading graphs:

Description: Graphs will be shown to the user, providing visual representations of information such as price action in multiple time frames, Profit and Losses, candle stick charts and line charts.

Criticality: This requirement is of high criticality, as users must be able to have access to a visual representation of each currencies price action.

Technical issues: Graphs must be able to display data accurately and efficiently due

users basing their trades off the information provided. Google charts and Trading view Api will be used to implement the graphs.

Dependencies with other requirements: Users must have an account and be logged in to access their data and view the charts.

Analysing Ai newsfeed:

Description: Ai analysis will be provided for each news post that appears in the news feed. This analysis will consist of a short summary of the news and a bullish, bearish or neutral sentiment

Criticality: This requirement is of high criticality, as users will need to be able to interpret the news fast to see how it affects their current trades or future trades

Technical issues: Ai will be trained using a ml model, so there could be misinterpretations
if the ai isn't trained on enough data which could mislead the user, however a disclaimer will be provided to ensure users read the analysis with a pinch of salt.

Dependencies with other requirements: Users must have an account and be logged in to access the news feed.

Tracking Trader positions:

Description: Users will be able to view live trades from the top traders on Binance to have a better understanding of what currencies the best traders are trading and how they trade.

Criticality: This requirement is of low criticality, as top traders may have many different trading strategies that may not correlate with the users trading style, however, users can still have the option to view the live positions as they please.

Technical issues: We must effectively scrape the Binance leader board page accurately
using the Binance api and provide clear and up to date information to the user to avoid any confusion.

Dependencies with other requirements: Users must have an account

and be logged in to view the traders positions.

Ai chat bot querying:

Description: Users will have access to an AI powered chatbot that can answer any questions the user may have about trading.

Criticality: This requirement is of high criticality, as beginner users may have a lot of questions about trading and the markets.

Technical issues: We have to utilise the ChatGPT Api to be able to answer questions effectively and we may have to hard code some answers about our platform that the ChatGPT large language model may not be able to answer.

Dependencies with other requirements: Users must have an account and be logged in to access the chat bot.

Technical Indicators:

Description: Users will be able to use basic technical indicators such as RSI, MACD, Bollinger Bands and EMA which will provide more in depth information about the price action of a currency.

Criticality: This requirement is of medium criticality, as only the more advanced users will understand how to use the indicators.

Technical issues: We will have to use an API from trading view or Binance to pull the indicator data for each mock currency we have. The data will need to be accurate to avoid traders from misunderstanding the price action.

Dependencies with other requirements: Users must have an account and be logged in to access the technical indicators and they will have to use the trading graphs.

4. System Architecture

Frontend (HTMX, HTML, CSS)

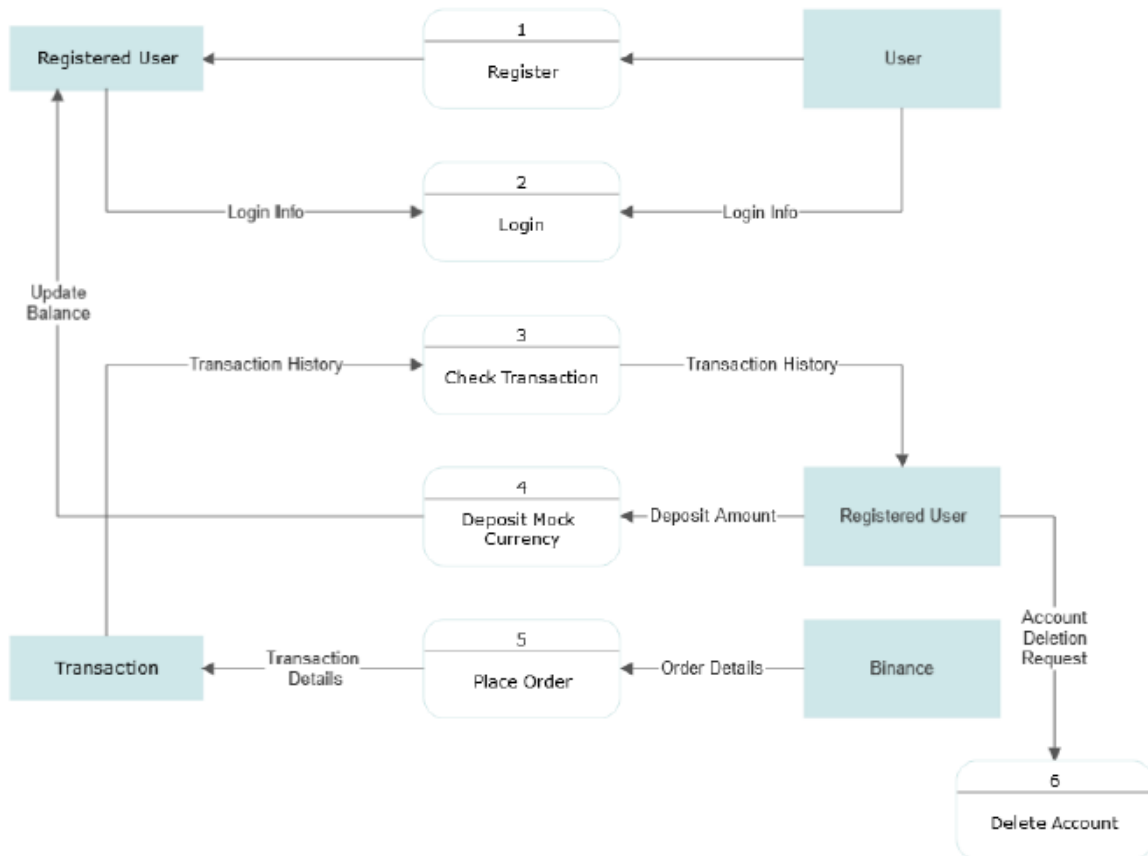
- **HTML/CSS:** These are fundamental web technologies used for structuring and styling the user interface of the trading simulator. HTML defines the structure of the web pages, while CSS provides styling for a visually appealing and user-friendly interface.
- **HTMX:** This is a library that will allow us to manipulate the DOM (Document Object Model) and make AJAX requests easily. In the context of the trading simulator, HTMX could be used for dynamic updates on the frontend without a full page reload, providing a smoother user experience. This will be useful when implementing the live news-feed, on which the AI will provide sentiment analysis.

Backend (Django):

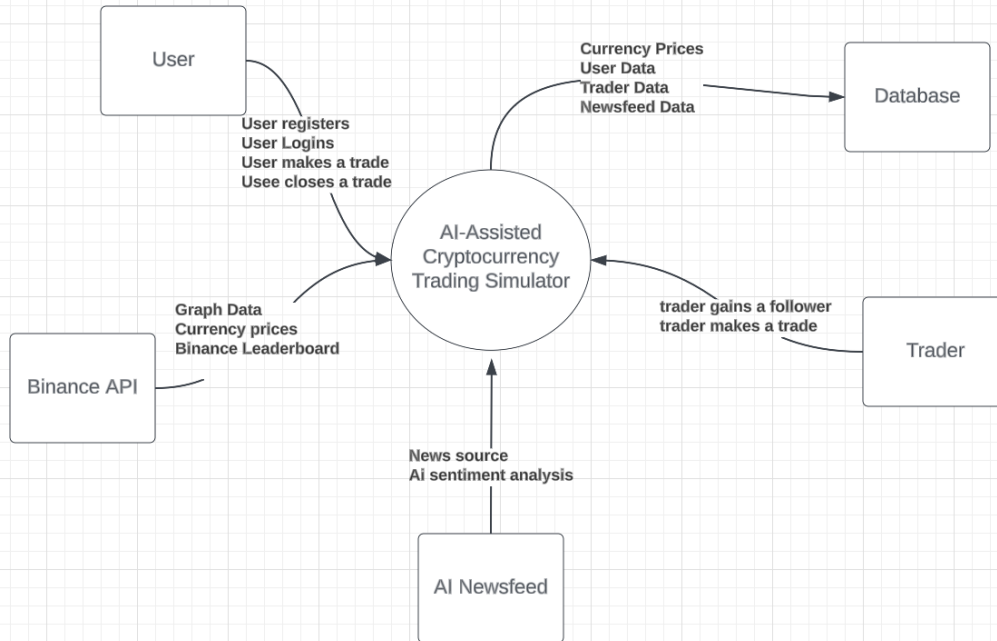
- **Django Framework:** Django is a high-level web framework for Python, known for its simplicity and versatility. It provides an MVC (Model-View-Controller) architecture, which we will use in organising code and separating concerns.
- **Binance API Integration:** We are integrating the Binance API to fetch real-time cryptocurrency data. This could include market prices, trading pairs, historical data, and other relevant information necessary for the trading simulator.
- **SQL Database:** Django supports various databases through its ORM (Object-Relational Mapping) system. We will use SQL databases (SQLite) to store user data, transaction history, and other persistent information related to the simulator.

5. High-Level Design

5.1 Data Flow Diagram

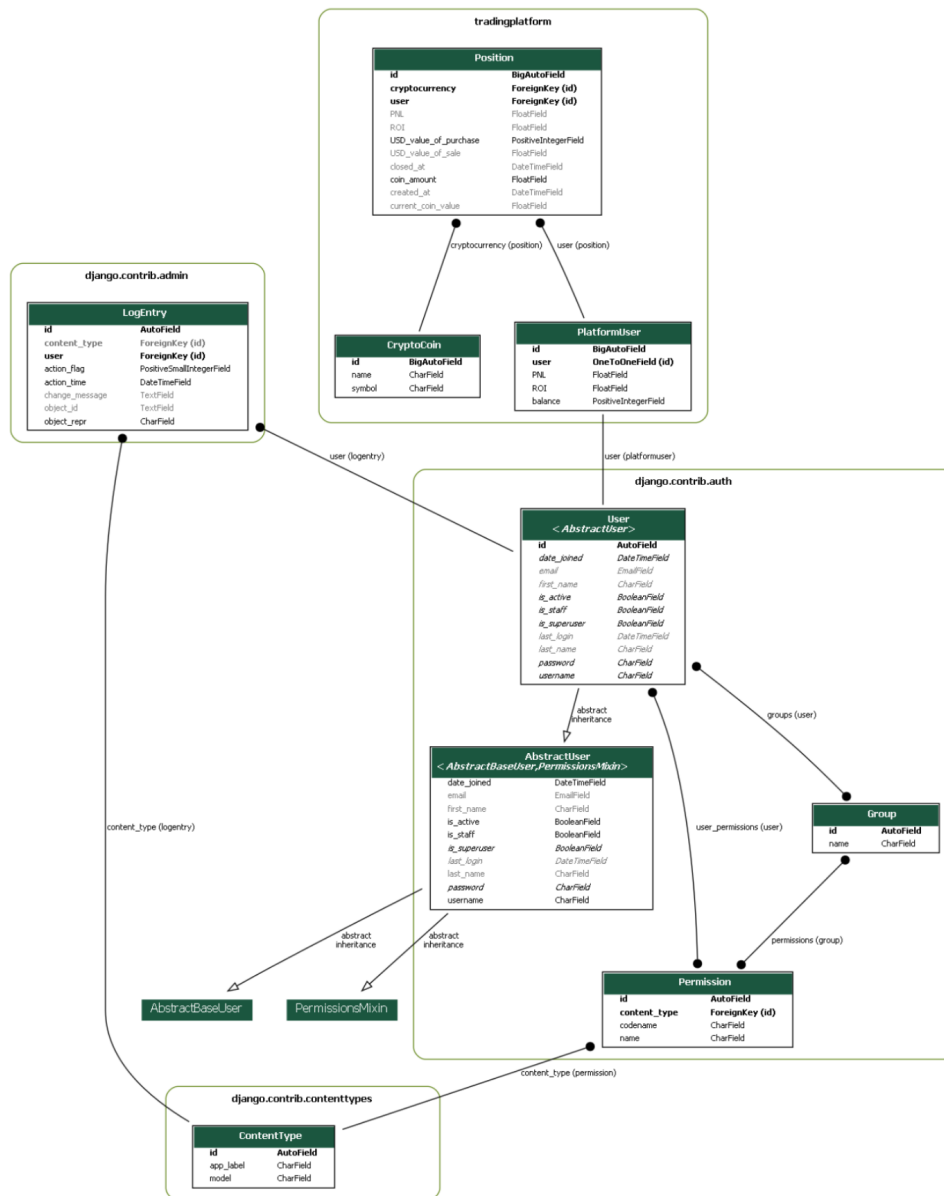


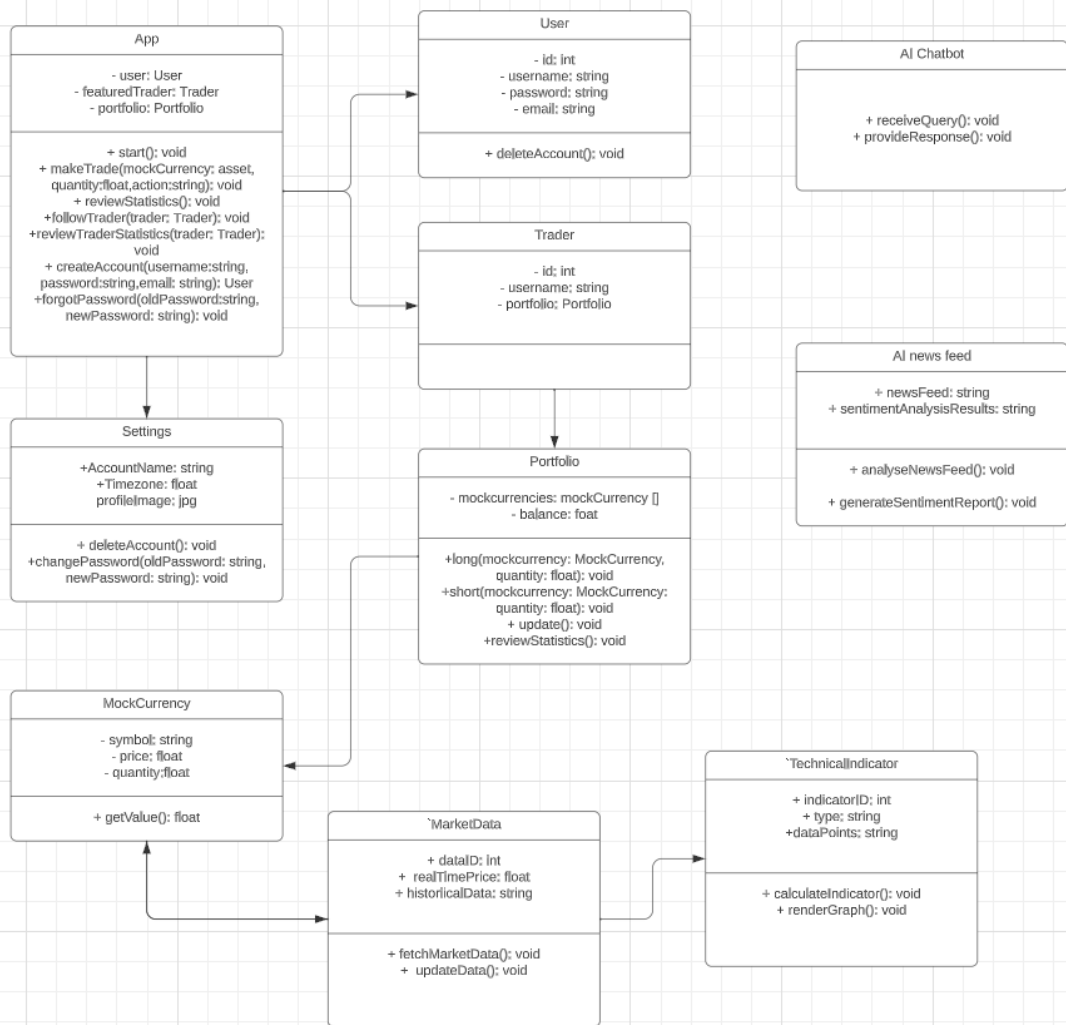
5.2 Context Diagrams



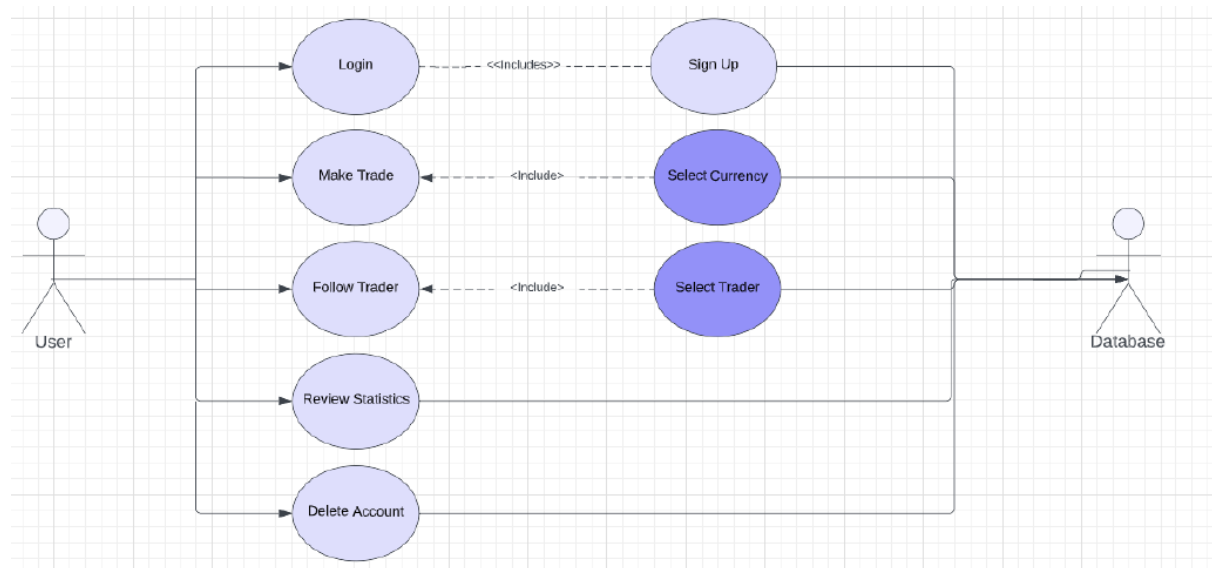
5.3 Class Diagram

Below is a UML class diagram for the Django models, showing the relation between the app and the User Authentication System.





5.4 Use Case Diagram



6. Preliminary Schedule

Task Name	Start Date	End Date	Duration (Days)
Functional Specification	01/11/2023	17/11/2023	17
Initialise Django Back-end	1/12/2023	09/12/2023	9
Christmas Exams Period	10/12/2023	22/12/2023	13
Finalise UI Design	26/12/2023	30/12/2023	5
Design AI model	01/01/2024	14/01/2024	14

UI Implementation	15/01/2024	21/01/2024	7
AI Implementation	22/01/2024	11/02/2024	21
Testing and Debugging	12/02/2024	25/02/2024	14
Technical Spec	26/02/2024	03/03/2024	7
Video Walkthrough & User Guide	04/03/2024	06/03/2024	3
Project Deadline	21/04/2024	21/04/2024	-
Project Demonstration	TBA	TBA	1

7. Appendices

Cryptocurrency News Feed

<https://news.treeofalpha.com/>

Binance Marketplace

<https://www.binance.com/en>

Binance Leaderboard API

<https://rapidapi.com/DevNullZero/api/binance-futures-leaderboard1>

Binance Api Documentation

<https://binance-docs.github.io/apidocs/futures/en/#change-log>

Ai chatbot API

<https://openai.com/blog/openai-api>

Trading view API

<https://www.tradingview.com/rest-api-spec/>

Google charts

<https://developers.google.com/chart>